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Integrating literature review into a conceptual framework for
monetizing water reuse benefitsEmmanuel Padmore Mantey^{ID} and Lu Liu^{* ID}

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^{*} Author to whom any correspondence should be addressed.E-mail: luliu@iastate.edu**Keywords:** water reuse, water reclamation, water recycling, monetization of environmental benefits, economic valuation, environmental externality, contingent valuationSupplementary material for this article is available [online](#)

Abstract

Water crisis, driven by scarcity, pollution, and uneven distribution, threatens water security worldwide. Water reuse projects can contribute to sustainable water-management objectives by augmenting water supplies, reducing pressure on freshwater sources, and generating economic, environmental, and social co-benefits when implemented within appropriate hydrological, institutional, regulatory, and governance contexts. However, prevailing evaluation approaches often remain incomplete, emphasizing direct financial gains (e.g. avoided imports and cost savings) while undercounting broader benefits such as energy savings, greenhouse gas reductions, water quality improvements, and flood mitigation. This review addresses these gaps by synthesizing the full spectrum of quantifiable benefits associated with water reuse, organized across three dimensions: (i) economic valuation approaches, (ii) benefit categories spanning water supply, water quality, and flood mitigation, and (iii) sources of water to be reused. Following PRISMA guidance, we screened 7592 records retrieved from Web of Science and included 110 sources (74 peer-reviewed papers and 36 reports) that quantified or qualitatively evaluated reuse benefits. Results show that more than half of the included studies prioritize water supply benefits, while benefits tied to water quality and flood mitigation are comparatively underexamined and inconsistently monetized. Based on the evidence synthesis, we propose a conceptual framework that organizes water reuse benefits into major pathways and links them to measurable indicators, valuation methods, and aggregation considerations. We conclude by outlining research and practice priorities, including the systematic incorporation of water reuse benefits into economic evaluations to strengthen benefit accounting and support decision-makers in maximizing the value of water reuse investments.

1. Introduction

The water crisis, marked by scarcity, pollution, aging infrastructure, climate variability, and uneven distribution, poses an urgent threat to sustaining life and economic growth [1–4]. These pressures affect both water quantity and quality, creating management challenges that cannot be addressed solely through supply expansion. In many regions, growing municipal, agricultural, industrial, and environmental water demands are occurring alongside declining freshwater reliability and increasing treatment needs.

These conditions have intensified interest in alternative water sources that can augment existing supplies, reduce pressure on freshwater systems, and support more resilient water-management portfolios [5]. In the United States, for example, water scarcity in the Colorado River Basin is intensifying due to decades of chronic overuse and prolonged drought, resulting in millions of dollars in municipal and recreational losses across Arizona, Nevada, and California [6–9]. Water quality issues compound these shortages: agricultural runoff, industrial discharge, and urban stormwater degrade freshwater systems [10].

Aging infrastructure further strains the system, as millions of deteriorating pipes leak trillions of gallons of treated water and pose contamination risks [11, 12]. Together, these quantity and quality challenges underscore the urgent need for cooperative, innovative, and sustainable water management solutions.

Water reuse, also referred to as water recycling or water reclamation, has emerged as a key strategy to address these challenges by reclaiming water from municipal, industrial, or environmental sources and treating it for beneficial use [13]. Potential source water for reuse may include stormwater, agricultural return flows, industrial process water, and municipal wastewater [14, 15]. Reuse can be unplanned, as when downstream communities withdraw from rivers containing upstream treated wastewater, or planned, where systems are intentionally designed for reuse [16, 17]. U.S. reuse practices follow a ‘fit-for-purpose’ principle, tailoring water quality to intended use, and are commonly classified into direct potable reuse (DPR), indirect potable reuse (IPR), and non-potable reuse (NPR) [18, 19]. DPR involves advanced treatment of reclaimed water to potable standards before its direct introduction into municipal systems [20]. IPR introduces treated water into an environmental buffer, such as an aquifer or reservoir, before further purification [21, 22]. NPR uses treated water for non-potable purposes such as irrigation, cooling, or landscaping.

The economic, environmental, and social benefits of water reuse are increasingly recognized as important contributions to broader water-management objectives. Although water reuse is primarily a supply-augmentation strategy, it can support broader sustainability goals when integrated with other planning, governance, regulatory, and watershed management practices. Economically, reuse can lower the costs of water supply by reducing reliance on expensive or energy-intensive alternatives such as desalination and long-distance water transfers [23]. Environmentally, fit-for-purpose water reuse alleviates pressure on rivers and aquifers by reducing freshwater withdrawals, helping maintain ecological integrity and biodiversity [24–26]. Socially, water reuse enhances community resilience and well-being by providing a dependable alternative where conventional supplies are constrained or vulnerable [24, 25].

Water reuse projects around the world demonstrate a wide range of applications and co-benefits, such as stormwater reuse for flood mitigation in Australia [27, 28], rainwater reuse for domestic and commercial functions in Greece and Sweden [29–31], and the use of reclaimed water for agricultural irrigation in Mediterranean and Middle Eastern countries [32–34]. In the U.S., the Environmental Protection Agency (EPA) promotes water reuse through the National Water Reuse Action Plan (WRAP), which coordinates with diverse stakeholders to address technical, institutional, and financial barriers. WRAP

emphasizes optimizing water reuse at the community scale before returning water to the environment, thereby maximizing overall water efficiency [35]. In addition, the EPA’s Clean Water State Revolving Fund supports water reuse projects by highlighting and financing effective state policies and practices that encourage sustainable water management [36]. NPR applications such as landscape irrigation, industrial processes, and cooling water are widely practiced in many U.S. states [19]. The adoption of potable reuse remains more uneven. IPR has been implemented in several states, particularly in water-scarce regions such as California and Texas, whereas DPR remains relatively limited [37–39], constrained by high capital and operating costs [40–42]. Florida employs reclaimed water for environmental restoration, industrial cooling, and aquifer recharge [40]. While in Texas, El Paso has emerged as a national leader in DPR through the development of the Pure Water Center, supplementing drinking-water supplies in an arid region through advanced purification of treated wastewater [41, 42].

Research to date has demonstrated the technical feasibility of various treatment methods for reclaimed water [43], the potential for reuse to reduce freshwater withdrawals [44, 45], and the success of policy-driven adoption in certain U.S. states [45, 46]. However, the benefit valuation literature remains fragmented. Many assessments focus on direct financial outcomes, such as avoided imported water costs, avoided water purchases, or operating-cost savings, while broader environmental and social co-benefits are often treated qualitatively or excluded. Benefits such as water-quality improvement, energy and greenhouse gas effects, flood mitigation, ecosystem enhancement, and community resilience are particularly difficult to compare because they occur across different spatial scales, time horizons, and institutional boundaries. This creates a benefit-monetization problem: reuse projects may appear costly when individual benefits are assessed, but they may provide additional value when these benefits are integrated into a singular framework. At the same time, including these benefits without clear definitions, baselines, and valuation boundaries can overstate project value or lead to double counting.

In this literature review, we address these gaps by synthesizing findings from 110 qualifying studies, categorizing water reuse benefits across multiple dimensions. Although governance and regulatory conditions are important contextual factors because they shape project feasibility, implementation pathways, and the types of benefits that can be realized, they are not the primary focus of this review. Drawing from this review, a comprehensive cost–benefit framework was proposed to capture the full suite of reuse benefits. The resulting framework aims to equip decision-makers with a more complete valuation of water reuse, supporting more informed, transparent, and

financially sound water management strategies, with a particular emphasis on applications within the U.S.

2. Methods

A systematic search of documents was conducted in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, as outlined by Moher *et al* [47]. This effort aimed to scrutinize peer-reviewed journals that address existing methods for monetizing economic, environmental, and social benefits of water reuse, as well as studies assessing the use and associated benefits of various water sources intended for reuse. PRISMA was used as a reporting and screening framework rather than as a statistical meta-analysis protocol, because the included studies varied substantially in source water, end use, valuation method, spatial scale, and benefit category, providing a transparent and reproducible structure for analysis. To maintain the highest quality of our assessment, we restricted ourselves to sources that provided methodologically traceable evidence on the benefits of water reuse or on valuation approaches. Peer-reviewed journal articles were prioritized because they generally provide transparent methods, empirical analysis, and documented assumptions. However, peer-reviewed literature alone does not fully capture the evidence base for water reuse benefit valuation, because many project-level benefit estimates are produced by federal or state agencies, water boards, municipal utilities, regional water-management agencies, and public institutions involved in implemented or planned water reuse projects. Therefore, official reports from government agencies and credible public institutions were included when they contained clearly documented valuation assumptions, project-specific benefit estimates, or methods relevant to water reuse benefit accounting. By contrast, editorials, letters, news articles, webpages lacking methodological detail, non-authoritative grey literature, and documents focused solely on technical design without benefit evaluation were excluded because they did not provide a sufficient analytical basis for coding or comparison. The comprehensive search began in August 2022 and ended in May 2024, leveraging the extensive Web of Science database. The search strategy, delineated in table 1, employed a combination of three core aspects:

1. Economic Analysis: This aspect investigates the diverse monetization methods formulated or adopted globally, with a significant focus on practices within the U.S. The goal was to understand the economic implications and effectiveness of these methods in different contexts.
2. Benefits of Water Reuse: This aspect examines the advantages associated with reusing water,

Table 1. Core aspects of literature search and associated keywords.

Economic analysis
1. Monetization of water reuse
2. Economic valuation
3. Contingent valuation OR Willingness to pay OR perception
4. Ecological cost OR benefit
5. Conjoint analysis
6. Acceptability OR Willingness to accept
Benefits of water reuse
7. Environmental benefits
8. Cost and benefits
9. Benefit estimation
10. Recreational benefit
11. Water quality OR savings
12. Water externality OR externality
13. Water reuse OR water recycling OR water reclamation
Sources of water
14. Wastewater
15. Stormwater
16. Rainwater

highlighting the economic, environmental, and social benefits. It includes an analysis of case studies and practical implementations of water reuse systems.

3. Sources of Water for Reuse: This aspect explores the various sources of water that can be repurposed for reuse.

The detailed methodology, search parameters, and the keywords and search phrases used in the literature search are presented in table 1. The core aspects were combined using Boolean operators. Within each core block, related terms were connected by OR, while the three core aspects were connected by AND.

The combined search string used in the review for both peer-reviewed journal articles and government report includes: (('water reuse' OR 'water recycling' OR 'water reclamation' OR 'reclaimed water') AND ('wastewater' OR 'stormwater' OR 'rainwater' OR 'agricultural runoff') AND ('monetization' OR 'economic valuation' OR 'contingent valuation' OR 'willingness to pay' OR 'willingness to accept' OR 'perception' OR 'conjoint analysis' OR 'ecological cost' OR 'ecological benefit' OR 'environmental benefit*' OR 'cost benefit*' OR 'cost–benefit*' OR 'benefit estimation' OR 'recreational benefit*' OR 'water quality' OR 'water saving*' OR 'water externalit*' OR 'externalit*'))

This search string represents the overall Boolean logic used to structure the search. In practice, the search was implemented iteratively using combinations of the three concept blocks shown in table 1. This approach was used because a single highly restrictive search string can exclude relevant studies

Table 2. Inclusion and exclusion criteria for the systematic review.

Category	Included	Excluded
Source of water	Wastewater, stormwater, and rainwater	Any source that did not relate to any of the sources mentioned except agricultural runoff
Methodology	Quantitative and qualitative documents that evaluate or report on water reuse benefits, monetization of environmental externalities	Documents that did not specify any input on the monetization of environmental externalities, unclear methodologies, insufficient data, and papers that focus on the technical & design aspect
Document types	Published journal article, published conference proceedings, and reports (Credible organizations and government agencies)	Editorials, letters, grey literature, news reports
Publication language	English	Non-English

that discuss water reuse benefits using different terminology.

Table 2 presents the full set of inclusion and exclusion criteria used in our review. The criteria cover three primary categories of water sources for reuse: wastewater, stormwater, and rainwater. While limited qualitative information on the benefits of agricultural runoff reuse was considered, studies were excluded if they did not include quantitative evaluations related to agricultural runoff. We also excluded documents lacking a clear methodology as well as those based on survey data with fewer than 100 responses. Finally, studies that only mentioned environmental externalities, without providing applied or methodologically grounded analyses, were not included. The report search was conducted as a targeted supplementary search to capture applied, project-level evidence that is often not published in peer-reviewed journals. Government reports were retained only if they met the same substantive eligibility criteria applied to academic literature: they had to evaluate or report water reuse benefits, provide a clear methodology or documented assumptions, and include quantitative benefit estimates or information relevant to benefit monetization. Reports were excluded if they were news items, editorials, opinion pieces, advocacy materials, webpages without methodological detail, technical design documents without benefit evaluation, or documents that only mentioned reuse benefits without providing analytical support.

Using this targeted procedure, 40 reports were assessed for eligibility, of which 36 were included in the final review documents. The inclusion of these reports improved coverage of applied and project-level valuation evidence, particularly for U.S. water reuse and stormwater capture projects.

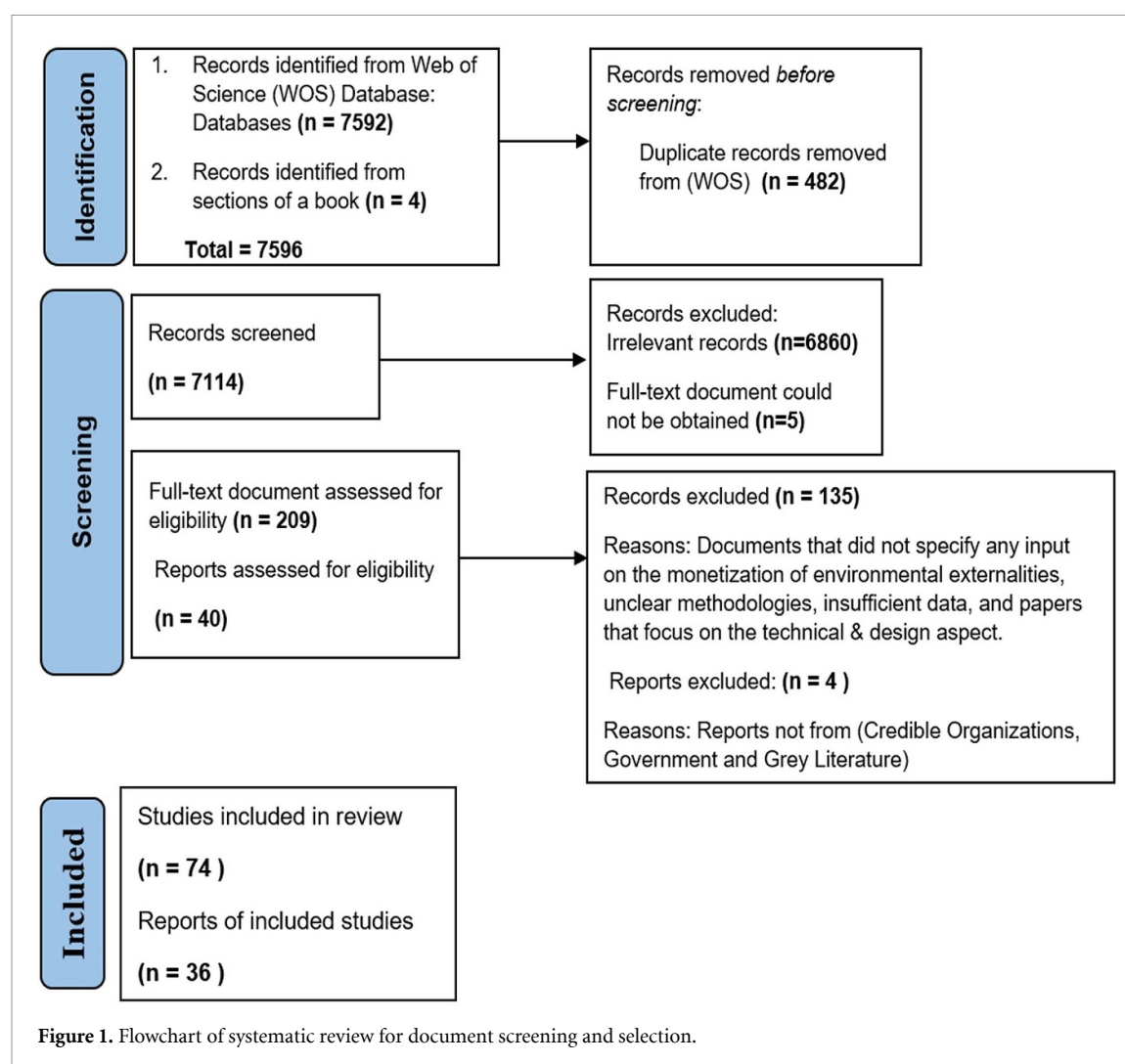
2.1. Review process

We conducted a thorough search of the Web of Science database, yielding 7592 sources, which included peer-reviewed papers, conference papers, and four book sections, totaling 7596 documents. After eliminating duplicates, we were left with

7114 unique records. By reviewing titles and abstracts, we found 6860 records to be irrelevant and could not access the full texts of five articles. This narrowed our pool to 209 documents for a more detailed full-text review.

Following the full-text review, we determined that 135 of these documents were irrelevant, leaving us with 74 documents for our analysis. These consisted mostly of journal articles (67), along with conference proceedings (3) and book sections (4). Additionally, we incorporated 36 reports from U.S. government agencies (i.e. California Department of Water Resources, State Water Resources Board, City of Fresno, City of Fontana, etc) that have implemented reuse projects. Our screening process yielded 110 documents for the final analysis (figure 1). Documents were first classified according to whether they contained at least one quantified benefit evaluation. A document was classified as quantitative if it applied a method, metric, or model to estimate, monetize, or numerically evaluate at least one water reuse benefit. Examples include avoided imported water costs, avoided treatment costs, willingness to pay, avoided flood damage, nutrient recovery value, energy or greenhouse gas reductions, and other quantified benefit measures. Using this definition, 78 of the 110 reviewed documents contained at least one quantitative benefit evaluation.

A document was classified as qualitative-only if it discussed water reuse benefits, benefit pathways, or valuation methods without producing a quantitative or monetary estimate for any benefit. Using this definition, 32 of the 110 reviewed documents were qualitative-only. However, qualitative findings were summarized at the benefit level. Therefore, a document that quantified one benefit could also be coded qualitatively for another benefit if that additional benefit was only discussed or identified without being monetized or numerically evaluated. For this reason, 88 documents contained at least one qualitative-coded benefit or method discussion, even though only 32 documents were qualitative-only. For this review, papers refer to peer-reviewed studies,

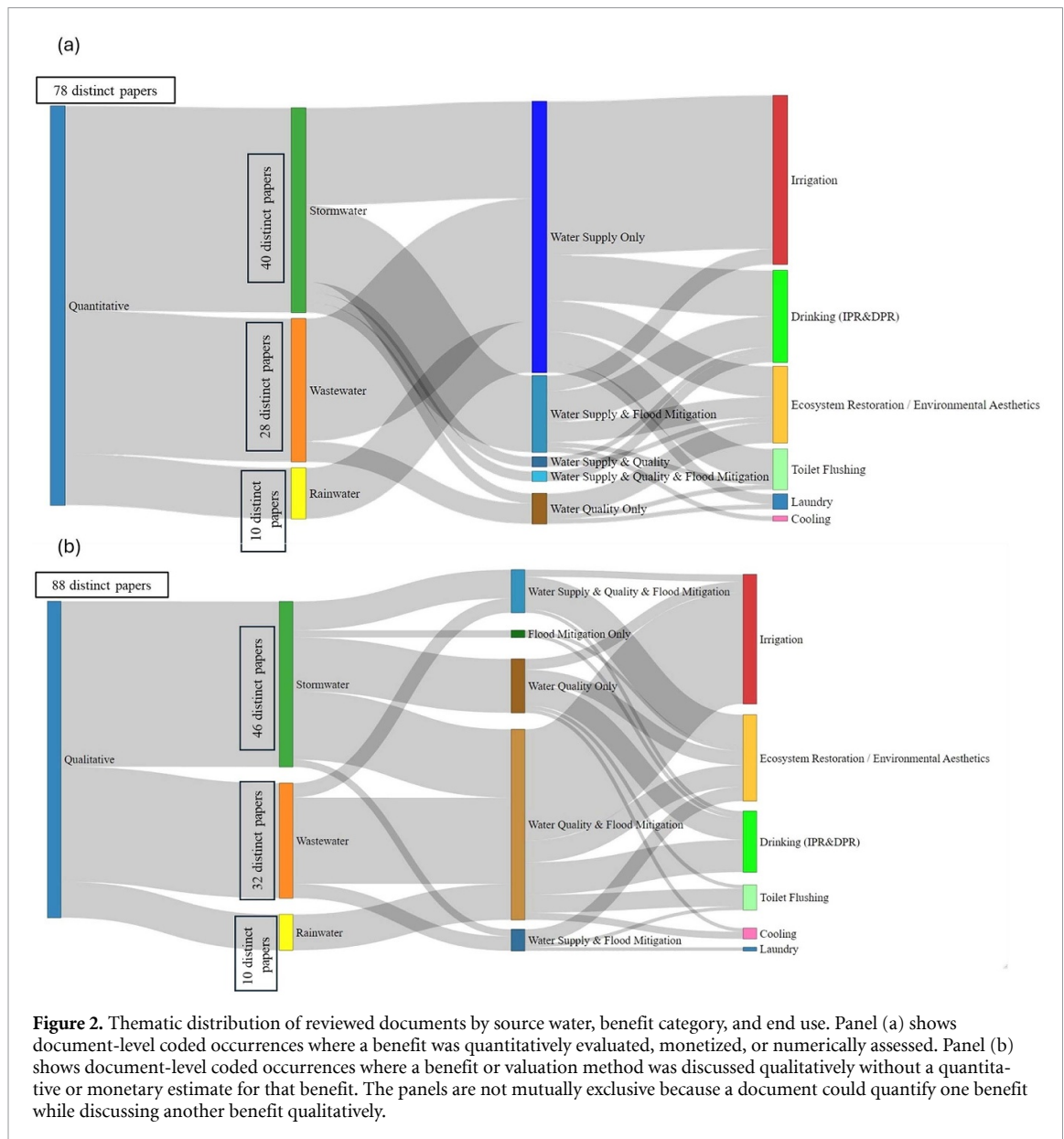


reports refer to credible government reports, and documents refer to the final scope of papers and reports used for analysis.

To improve transparency and reproducibility, the complete review corpus is provided as a separate supplementary Excel file, Supplementary Dataset S1. This dataset contains a document-level coding matrix for all 110 included documents, including the 74 peer-reviewed sources and 36 agency/government reports. For each document, the matrix reports the author or organization, year, document type, study location, geographic category, source water or reuse application, benefit category, valuation or assessment method, quantification status, and role in the synthesis. This dataset allows readers to verify the geographic composition of the evidence base and the classifications used to distinguish quantitative, qualitative-only, peer-reviewed, and report-based evidence. Supplementary table S2 is retained as a focused synthesis table because it serves a different purpose: rather than representing the complete corpus, it summarizes selected studies that directly informed the economic evaluation and valuation-method discussion.

3. Results and discussion

In this review, the 110 documents were categorized as either quantitative or qualitative. Figure 2 illustrates the thematic distribution of studies examining the benefits of water reuse, including the source water, the types of benefits assessed, and the intended end uses. The height of each bar reflects the number of document-level coded occurrences for each benefit category and should be interpreted as the frequency with which that benefit appeared in the review corpus. It is found that most research centers on stormwater reuse, followed by wastewater reuse, with rainwater reuse emerging as the least studied area. This imbalance in research attention underscores the need for more targeted studies on rainwater reuse. The benefit categories examined in the reviewed studies, as depicted in figure 2, primarily include water supply, water quality, and flood mitigation. Several documents monetized more than one benefit, which is reflected in the aggregated categories shown in the figure. Notably, only a limited number of studies focus exclusively on water quality benefits, highlighting an important area in need of further research.



In terms of end-use applications, most studies center on irrigation, drinking water (IPR and DPR), and ecosystem restoration, while considerably fewer address uses such as toilet flushing, laundry, and cooling, indicating a narrow application spectrum that leaves substantial opportunities for further investigation. Within the IPR/DPR category, more than 80% of studies concentrate on IPR. However, DPR regulations have recently begun to advance in states such as Colorado, California, Arizona, and Texas, suggesting growing momentum and the need for additional research in this area [48]. It is worth noting that 55 out of the 110 studies are U.S. based and the remaining studies represent Europe and other regions. Although the U.S. literature constitutes a substantial share of the evidence base, the geographic diversity of the reviewed studies suggests that the findings are broadly relevant across a range of water reuse contexts.

3.1. Existing monetization methods

To improve the transparency and reproducibility of the quantification and monetization of water reuse benefits, we developed a table (table S2) listing the documents that directly informed the synthesis of monetization methods. For each document included in this table, we report the author and year, the benefit or outcome measured, the quantification or valuation method used, the method's application, and the most appropriate application context (table 3).

Economic valuation and monetization methods are central to capturing both the market and non-market benefits of water reuse, extending traditional financial assessments to include environmental and social co-benefits. The literature identifies seven principal approaches: stated preference methods, revealed preference methods, cost-benefit analysis (CBA), life cycle assessment (LCA) and life-cycle cost

Table 3. Detailed monetization-method table for the quantitative documents: what it measures, method, how it works, and ideal application.

Method	Authors	What it measures	How it works	Ideal application
Cost–Benefit Analysis (CBA)	Halaburka <i>et al</i> [77], Dallman <i>et al</i> [78], Diringier <i>et al</i> [38], Domènech and Sauri [79], Folkins <i>et al</i> [80], de Gois <i>et al</i> [81], Chen and Wang [82], de Simone Souza <i>et al</i> [83], Declercq <i>et al</i> [84], Alfranca <i>et al</i> [85], The City of Calimesa report [38], The City of Somona county report [38], The City of Santa Margarita report [38], The City of Orange County report [38], The City of San Marcos report [38], The City of Palmdale report [38], The City of Menifee report [38], The City of Fresno report [38]	Net economic value of projects	Compares total costs vs total benefits over time	Comparing project alternatives by monetizing costs and benefits over time.
Life Cycle Assessment (LCA) & Life Cycle Cost Analysis (LCCA)	Wang and Zimmerman [86], Ghimire <i>et al</i> [61], Cooley <i>et al</i> [87], Wang <i>et al</i> [88]	Environmental impacts & Long-term financial costs	Evaluates impacts across full system lifecycle & Includes capital + operation costs over time	Good for financial planning.
Stated Preference (WTP)	Nelson <i>et al</i> [89], Verlicchi <i>et al</i> [90], Hurlimann and McKay [91], Loomis <i>et al</i> [92], Alcon <i>et al</i> [93], Kontogianni <i>et al</i> [94], Tziakis <i>et al</i> [95], Dupond <i>et al</i> [96], Menegaki <i>et al</i> [97], Petousi <i>et al</i> [98], Abu Madi <i>et al</i> [99], Lazaridou <i>et al</i> [100], Hurlimann and McKay [101], Frew <i>et al</i> [102], van Eeden <i>et al</i> [103], Martínez-Españeira [104]	Non-market values (e.g. water quality, recreation)	Surveys to estimate willingness to pay	Non-use values like preserving ecosystem.
Revealed preference (hedonic)	Halaburka <i>et al</i> [77], Kazmierczak [105], Knoche and Lupi [106]	Behavioral economic value	Infers value from observed market	Property price premiums near water bodies.
Benefit transfer	Halaburka <i>et al</i> [77], Molinos-Senante <i>et al</i> [107]	Estimated value from prior studies	Applies existing valuation to new context	Preliminary cost-effective and quick.
Production function approach	Bellver-Domingo <i>et al</i> [108], Molinos-Senante <i>et al</i> [62], Hernández-Sancho <i>et al</i> [109], Molinos-Senante <i>et al</i> [110]	Economic impact of environmental change	Links environmental inputs → economic outputs	Avoided agricultural fertilizer costs
Multi-criteria decision analysis (MCDA)	Das and Radhakrishnan [111], Hadipour <i>et al</i> [112], Paul <i>et al</i> [113]	Multi-dimensional decision value	Combines economic, environmental, social criteria	Handles trade-offs well

analysis (LCCA), benefit transfer, production function approaches, and multi-criteria decision analysis (MCDA). Together, these methods provide a broad analytical foundation for evaluating the economic viability of water reuse projects, though each has distinct strengths and limitations. In this review, environmental benefits are defined as measurable positive changes in environmental conditions resulting from water reuse or a water reuse project relative to a clearly specified baseline, i.e. a case without water reuse. These benefits differ from broader environmental valuation exercises because they are limited to outcomes that are directly linked to the reuse pathway, project design, and displaced alternatives and have been

categorized into 3 sections, namely: water supply, water quality, and flood mitigation. Environmental benefits may include reduced freshwater abstraction, reduced wastewater or stormwater discharge, lower pollutant and nutrient loads to receiving waters, groundwater recharge, streamflow or ecosystem management, avoided energy use and emissions where reuse displaces more energy-intensive supply options, and reduced runoff-related impacts where stormwater is captured and reused. Environmental externalities are the positive or negative environmental effects of reuse that are not fully reflected in project financial accounts, but that affect ecosystems, downstream users, or society more broadly. Monetization

is therefore appropriate only when these effects can be linked to measurable biophysical change and valued against an existing condition.

Stated preference methods, including contingent valuation and choice experiments, elicit willingness to pay for environmental improvements under hypothetical scenarios, allowing the valuation of both use and nonuse benefits when market data are unavailable [49–52]. These methods are well suited for assessing water quality improvements and ecosystem services but are sensitive to survey design and hypothetical bias [53, 54]. Revealed preference methods, such as hedonic pricing and travel cost models, infer values from observed behavior, for example, property price premiums associated with improved water quality or recreation demand for enhanced water bodies [55–58]. While grounded in actual choices, these methods typically capture only use values and require extensive data and strong modeling assumptions [54, 59].

Traditional CBA, LCA, and LCCA complement these valuation tools. CBA monetizes costs and benefits over time but often struggles to incorporate non-market environmental and social impacts. LCA provides a system-wide assessment of environmental burdens across the full life cycle of water reuse systems, while LCCA estimates long-term financial costs, including capital and operations [60, 61]. Although these approaches support comparative evaluation, they usually require additional non-market valuation techniques such as contingent valuation, choice experiments, hedonic pricing, benefit transfer, and avoided-cost methods to monetize ecological and social co-benefits such as improved water quality, ecosystem services, and public health benefit [62].

The benefit transfer method offers a cost-effective means of estimating non-market benefits by applying values from existing studies to new contexts, making it useful for preliminary assessments [63–65]. However, its reliability depends on close alignment between study and policy sites [66, 67]. Production function approaches link biophysical changes such as nutrient delivery or improved water availability to economic outputs in sectors like agriculture, thereby valuing ecosystem services through their contribution to production [68–71]. These methods provide clear ecological–economic relationships but require detailed data and generally omit cultural or non-use values [71, 72].

In recent years, MCDA has been increasingly applied to water reuse planning to integrate economic, social, and environmental criteria, enabling explicit evaluation of trade-offs across benefit categories [73, 74]. MCDA is a structured decision-support framework that evaluates and ranks alternatives by assigning weights to multiple, often competing criteria and aggregating their performance across dimensions. This is particularly important in water reuse systems, where benefits are multidimensional

and not directly comparable. However, many environmental impacts associated with water reuse lack observable market prices [75, 76]. To address this gap, non-market valuation techniques such as the directional distance function (DDF) have been employed. Rooted in production theory, the DDF quantifies the simultaneous expansion of desirable outputs and contraction of undesirable outputs (e.g. nitrogen, phosphorus, and biological oxygen demand), allowing for the derivation of shadow prices. These shadow prices represent the implicit economic value of reducing pollutants, thereby incorporating environmental externalities into economic evaluation frameworks [62].

We summarized monetization methods for the quantitative documents in table 3. This table clarifies the methodological basis for the synthesis and to distinguish papers that directly quantified or monetized benefits from those that primarily provided conceptual, policy, or qualitative discussion.

3.2. Monetizing benefits of water reuse

3.2.1. Water supply benefits

Water supply benefits defined in the literature largely focus on water savings [114, 115], defined as reductions in freshwater demand achieved by recycling and repurposing treated stormwater, wastewater, and rainwater. This could lead to avoided imported water cost [38, 77], which represents the financial savings realized when communities reduce dependence on expensive imported supplies. In areas with limited local resources, imported water often entails substantial transportation, infrastructure, and energy consumption. More than 50% of the documents reviewed ($n = 53$) explicitly focused on water supply benefits, of which more than 20 peer-reviewed documents quantified water savings or avoided imported water costs. However, estimates of imported water costs typically include only transportation expenses, assuming potable quality water, while omitting other influential parameters such as energy consumption and associated CO₂ emissions [38]. Fewer than eight documents quantified more than two benefit components. Lazaridou *et al* and Tziakis *et al* expanded their analyses to incorporate carbon emissions by applying the social cost of carbon [95, 100]. Additional important components, such as energy savings from reduced pumping (especially when groundwater is the primary source), were noted but infrequently quantified. However, emphasis on energy-related benefits should be evaluated as net benefits relative to the water-supply option or treatment pathway that water reuse replaces. Reclaimed water still requires treatment to meet fit-for-purpose standards; therefore, the energy and cost associated with producing reclaimed water should generally be counted as part of the project cost, not as an avoided-cost benefit. Energy savings occur primarily where reuse reduces dependence on more energy-intensive supply

options, such as imported water, long-distance conveyance, deep groundwater pumping, desalination, or emergency water-supply development.

Treatment-related savings should only be counted as benefits when reuse clearly avoids treatment that would otherwise be needed elsewhere in the system. Examples include avoiding additional advanced treatment at a drinking-water facility, reducing pollutant removal requirements before discharge, or lowering downstream treatment needed to meet regulatory or water-quality objectives [116]. Making this distinction is important because counting reuse-related treatment requirements as benefits could overstate the project's value and weaken the economic comparison between reuse and other supply alternatives.

Cooley *et al* (2019) and Diringer *et al* both quantified water supply benefits in the context of stormwater capture and reuse, using the levelized cost method [38, 87]. Both studies noted that some water supply options entail additional costs and benefits that are often insufficiently quantified, including environmental externalities (e.g. impacts on ecosystems and reduced surface water withdrawals), water quality improvements, avoided pollution and downstream treatment costs, flood risk reduction, and broader social co-benefits such as habitat enhancement and recreational value, and therefore could not be incorporated into their analyses [87].

Declercq *et al* reported that, in the Cannes case, treated wastewater reuse was evaluated as a local supply and recharge option against a business-as-usual scenario involving external water supply. This shows that reuse may not replace large-scale transfers on a one-to-one volume basis, but can still provide value by offsetting marginal demand, supporting seasonal supply reliability, and reducing dependence on external sources [84].

Other studies show how water supply and energy-related benefits depend on the avoided alternative. In the Ballona Creek watershed, rainwater harvesting was evaluated using modeled reductions in potable water demand and stormwater runoff, with additional monetization of energy and carbon savings [78]. This type of example shows that energy benefits should be linked to the specific treatment, pumping, conveyance, or supply pathway avoided by the reuse project.

Wastewater reuse for agriculture also demonstrated significant reductions in fertilizer demand, enabling valuation of avoided fertilizer costs alongside water savings [95, 100]. A representative example is the Bay Area Regional Recycled Water Program in California, which plans to deliver 1720 acre-feet per year of recycled water for non-potable uses across the region [38]. The program offers substantial economic benefits, primarily by avoiding the high costs of potable water, which averages \$1500 per acre-foot for retail suppliers. In addition, the recycled water contains valuable nutrients such as nitrogen,

phosphorus, and potassium, that are typically absent in potable water. The estimated combined commercial value is about \$30 per acre-foot [38].

Examples of agricultural reuse also show the importance of local context. In Crete, studies of reclaimed-water irrigation found that acceptance and willingness to pay varied by crop type, treatment level, perceived health risk, and trust in managing authorities [97, 98]. Farmers were more favorable toward reclaimed water for olive tree irrigation than for higher-contact crops such as vegetables or vines. These findings indicate that fertilizer savings, crop benefits, and irrigation cost savings should not be assumed to be uniform across agricultural settings; they depend on crop nutrient demand, water quality, irrigation practices, salinity risks, and farmers' willingness to use reclaimed water.

The 'willingness to pay' (WTP) method is also used to quantify water supply benefits [83, 95, 96, 100]. Recent work in the U.S. applied WTP approaches to evaluate the economic feasibility of water reuse systems. A contingent valuation survey of households in small community water systems was conducted and the estimated average WTP is approximately \$49 per month to support water reuse programs aimed at avoiding future water restrictions [52].

3.2.2. Water quality benefits

In this review, only eight documents explicitly focused on water quality benefits. The majority of the eight documents applied the WTP method, where wastewater was the primary source for reuse. In these cases, higher contaminant loads in discharged water negatively impacted ecosystems and recreational sites, which were essential for community relaxation and well-being [89, 90, 94]. For instance, a study in Australia assessed WTP for improved recycled water quality in homes, focusing on parameters such as salinity, color, and odor; due to water shortages, residents were willing to pay a premium for higher-quality water [91]. A study in California compared the costs of discharging secondary effluent into the ocean against discharging tertiary effluent (further treated by sand filtration or ultraviolet disinfection) into a small stream and wetland for ecosystem restoration. The analysis found that the benefits of discharging tertiary effluent outweighed the additional costs by \$1.8 million, with recreational value and scenic amenity providing the largest contributions [77].

Methods for monetizing water quality benefits in the reviewed literature are predominantly grounded in avoided cost frameworks, particularly within IPR and stormwater reuse applications [38]. In these approaches, improvements in water quality are not valued directly; instead, they are captured through cost savings from reduced treatment, chemical use, or infrastructure requirements. For instance, stormwater capture studies commonly quantify water quality

benefits as the avoided cost of treating polluted runoff prior to discharge or reuse, effectively substituting for conventional treatment systems. As demonstrated by Diringer *et al*, water quality benefits are monetized through avoided water treatment costs, along with other avoided expenditures such as regulatory compliance and downstream mitigation costs. These approaches implicitly treat improved water quality as a cost-saving mechanism, rather than valuing ecosystem or public health benefits directly, which remains less commonly quantified in monetary terms. Most studies evaluated non-potable uses, such as flushing toilets, cleaning bathrooms, laundry, and garden irrigation. These studies examined the quality of water entering ecosystems or recreational sites, with many of them originating from Europe [89]. For example, total habitat value for urban parks and aquatic ecosystems was derived from a meta-analysis of 39 wetland valuation studies, enabling a comprehensive assessment of ecological benefits [117]. Recreational use values, such as fishing, hunting, and wildlife observation, were assessed through survey-based valuation [38]. Furthermore, the avoided cost of sediment deposition was estimated by referencing reductions in sediment removal requirements for specific projects [38].

3.2.3. Flood mitigation benefits

Flood mitigation is included in the framework only where it is directly connected to stormwater capture, storage, infiltration, or reuse. It is not treated as a general benefit of all water reuse projects. Rather, it is a benefit pathway that can arise when stormwater reuse systems reduce runoff volume, delay peak discharge, lower pressure on drainage infrastructure, or reduce downstream flood exposure while also creating a reusable water supply. This distinction is important because potable reuse, wastewater reuse, and agricultural reuse do not necessarily provide flood mitigation benefits unless integrated with stormwater management. In this review, flood mitigation is therefore considered within scope only as part of multi-benefit stormwater capture and reuse systems.

Flood mitigation benefits are most directly linked to stormwater capture and reuse systems, rather than to water reuse projects broadly. These benefits occur when stormwater is intercepted, detained, infiltrated, or stored in ways that reduce runoff volume, delay peak discharge, lower flood depth, or lessen pressure on downstream drainage infrastructure. The benefit is therefore generated by the hydrologic function of capture and storage. Avoided flood damages may include reduced structural and contents losses for households and businesses, lower municipal cleanup and maintenance costs, reduced emergency response costs, and insured losses reflected in National Flood Insurance Program claims (NFIP) [38]. Also, it is important to highlight that stormwater capture projects have a strong economic benefit when

it provides multiple benefits at once, including water supply augmentation, water quality improvement, groundwater recharge, ecological enhancement, and avoided flood damages [118].

In this review, 17 documents focused on flood mitigation benefits. Various methods and parameters were employed across different studies. For instance, the avoided nuisance flood pumping costs were estimated by calculating the costs associated with emergency pumping efforts by the Los Angeles County Department of Public Works. This included the labor cost for two employees and a pumping truck, totaling \$3400 per event and \$6800 annually, given two annual flooding incidents [38]. Another flood-related benefit involves avoided city maintenance costs. For example, flooding in San Marcos Creek requires extensive cleanup efforts by the City of San Marcos, including deployment of public works crews costing approximately \$11 520 for an 11-person team. Additional contractor services, such as tree maintenance, cost about \$1320 per day, while equipment use and safety measures, such as debris trucks, loaders, barricades, and traffic control, add another \$3552. These costs illustrate the substantial financial burden that effective flood-control measures can help prevent.

The Flood Risk Assessment Model (FRAM) enables benefit-cost analysis of flood mitigation benefits. FRAM estimated damages using lump-sum values and focus solely on building and content losses while excluding impacts such as road damage, business interruption, or loss of life [38]. In contrast, the HEC-FDA model provided a more comprehensive assessment by accounting for structural and content damages across residential, commercial, and industrial buildings, as well as public spaces, vehicle damage, and post-flood cleanup [38].

Flood-control benefits were also evaluated using data from the NFIP, which provides insurance to property owners, renters, and businesses in participating communities. NFIP data indicate that, over the past decade, the average flood claim for both residential and commercial properties exceeded \$33 000. This value represents the mean damage per property per flood event and was used as a proxy for estimating avoided flood damages [38]. In practice, avoided damages are estimated by combining this average claim value with the number of structures exposed to flooding under different storm scenarios. For example, in the San Marcos Creek Floodway Improvement Project, hydraulic modeling identified between 126 and 182 structures at risk under 10 year to 100 year flood events. By applying the NFIP-based average damage value (\$33 000 per structure), total expected flood damages were estimated and combined with infrastructure losses (e.g. road damage) using models such as the Flood Rapid Assessment Model (FRAM). This resulted in an estimated annual expected flood damage of approximately \$698 000,

which, when projected over a 50 year project lifetime, yielded over \$11 million in avoided damages [38].

3.2.4. Social benefits of water reuse

Social benefits in water reuse refer to the positive impacts on communities and overall well-being [25]. These extend beyond direct economic and environmental gains to include improvements in public health, social equity, and community resilience. Water reuse can strengthen public health by lowering exposure to waterborne diseases in areas where untreated wastewater might otherwise contaminate local sources [119]. Water reuse can also promote social equity by providing more affordable water options for underserved communities and contribute to climate resilience by reducing vulnerability to drought and environmental stressors [119, 120]. Additional social benefits arise through support for recreational spaces and green infrastructure, as reused water can irrigate parks, gardens, and sports fields, improving urban quality of life.

Despite these substantial social benefits, most current methodologies do not fully capture them, resulting in their limited representation in cost-benefit analyses [121, 122]. Our review found that fewer than three studies explicitly incorporated social costs and benefits into their analytical frameworks. While several documents acknowledged the social advantages of reuse, there are few advanced methods for integrating them into comprehensive valuation approaches.

One methodology identified in our review, the net benefit value (NBV) model offers a more holistic framework for evaluating wastewater treatment and reuse, particularly in decentralized systems [82]. The NBV model expands beyond traditional cost-benefit analysis by incorporating both monetary and non-monetary factors across three categories: Natural Resources, Human Resources, and Human Health Protection. The Natural Resources category quantifies water conservation benefits based on local tariffs, capturing the economic value of reduced tap-water dependence. The Human Resources category evaluates labor requirements for operating and maintaining reuse systems and accounts for social costs by subtracting labor expenses while recognizing job-creation benefits. The Human Health Protection category estimates the economic value of reduced disease incidence using the human-capital method, linking health improvements to avoided treatment expenses and reduced productivity losses [82].

We summarized these findings in table 4 for the 78 quantitative documents and the 88 qualitative benefit-level observations.

The existing body of literature demonstrates a substantial methodological gap in the monetization of environmental and social benefits associated with water reuse projects [121]. In particular, current approaches often evaluate these benefits in isolation,

resulting in incomplete assessments that fail to capture the full range of economic, ecological, and social outcomes. To address these shortcomings, we recommend the adoption of an integrated conceptual framework that organizes benefits into three primary categories: water supply, water quality, and flood mitigation. The following sections outline recommendations for structuring this framework to support more comprehensive cost-benefit analyses.

3.3. Recommendations for integrated valuation

A benefit quantification framework is provided in figure S1 to clarify how water reuse benefits were organized into valuation pathways and linked to potential monetary indicators. The framework summarizes the main pathways for water supply, water quality, and flood mitigation benefits. The framework proposed in this review is not intended to be specific to the U.S. Although U.S.-based studies constitute a substantial portion of the reviewed literature, the review also includes evidence from Europe and other regions. Accordingly, the framework was designed to capture benefit pathways that are broadly applicable across diverse water reuse contexts and geographic settings, providing a general structure for organizing and assessing water reuse benefits.

3.3.1. Water supply benefits

First, we recommend expanding the valuation of water supply benefits beyond the commonly used metrics of water savings or avoided imported water, as seen in figure S1(a). A more integrated valuation should explicitly recognize the role of water reuse in augmenting and diversifying existing supply portfolios for both agricultural and non-agricultural users, particularly in regions facing increasing hydrologic variability and chronic water scarcity.

For agricultural applications, valuation frameworks should incorporate avoided fertilizer expenditures, as reclaimed water often contains nitrogen, phosphorus, and other nutrients that provide measurable agronomic value and can reduce the need for synthetic fertilizer inputs. These benefits are particularly relevant in nutrient-limited cropping systems and can represent a substantial portion of total economic value when reuse water is applied at scale.

For non-agricultural uses, such as cooling, toilet flushing, landscape irrigation, or industrial processes, we recommend explicitly differentiating between public and private water delivery systems, as their incentive structures, risk exposure, and cost recovery mechanisms differ substantially. Public water systems are responsible for maintaining reliable supply during droughts and periods of peak demand. In this context, water reuse can function as a drought-resilient supplemental supply, reducing reliance on costly emergency measures such as short-term water imports,

Table 4. Summary of benefit categories, valuation methods, reuse applications, and principal findings identified from the quantitative documents and qualitative benefit-level observations as described in section 3.2.

Evidence group	Benefits	Valuation methods	Reuse application	Principal findings
78 quantitative documents	Water supply, water quality, flood mitigation, energy, ecosystem services, recreation, property value, and social benefits	CBA, avoided-cost analysis, LCA/LCCA, stated preference, revealed preference, benefit transfer, production-function/shadow pricing, MCDA	Potable and non-potable reuse, irrigation, stormwater capture, groundwater recharge, wetland restoration, toilet flushing and industrial cooling	Quantitative studies most commonly monetized water supply and avoided-cost benefits. Ecosystem, recreation, and social benefits were less consistently monetized and often depended on proxy values or benefit transfer.
88 qualitative benefit-level observations	Reliability, resilience, public acceptance, ecological improvement, equity, water security, and avoided environmental degradation	Mostly conceptual discussion, qualitative assessment, or descriptive evidence	Broad reuse applications, including wastewater reuse, stormwater reuse, graywater, rainwater harvesting, and environmental reuse	Qualitative evidence broadened the range of recognized benefits but often lacked direct monetary estimates. These observations helped identify benefit pathways that are important but underrepresented in quantitative valuation.

intensified groundwater pumping, expedited development of new supply infrastructure, or the imposition of drought-related surcharges on consumers. By stabilizing supply availability, reuse projects can lower operational volatility, defer capital investments, and support more stable and predictable water tariffs over time. Private water delivery systems, including industrial self-supplied water users, private utilities, and facilities operating closed-loop or semi-closed-loop systems, face different economic incentives. For these entities, water reuse can reduce exposure to water market price volatility, supply interruptions, and regulatory risks, while simultaneously lowering costs associated with water procurement, wastewater discharge fees, and compliance with tightening effluent standards. In industrial settings, reuse can also enhance operational resilience by ensuring reliable on-site water availability, which is particularly valuable for continuous-process industries where downtime carries high economic penalties. Valuation frameworks should therefore capture not only avoided water purchase costs, but also risk-reduction and reliability benefits that are especially salient for private operators.

Finally, we recommend integrating energy-related benefits into the valuation of water-supply benefits by quantifying reductions in energy use associated with decreased reliance on imported water, long-distance conveyance, deep groundwater pumping, or energy-intensive treatment processes. These energy savings lead to corresponding reductions in associated CO₂ emissions, which can be monetized using estimates of the social cost of carbon. Incorporating this component captures broader societal benefits and provides a more comprehensive representation of the environmental performance of water reuse systems.

In addition, we recommend accounting for avoided opportunity costs associated with alternative water sources, including imported water, groundwater pumping, or surface water diversion. In water-stressed basins, these alternatives often impose high direct financial costs as well as indirect environmental and social burdens, such as aquifer depletion, land subsidence, ecosystem degradation, and reduced downstream availability. By displacing these marginal supplies, water reuse can generate value that is not captured by simple unit-cost comparisons alone.

3.3.2. Water quality benefits

We recommend conceptualizing water-quality improvements as a multidimensional and integrated benefit category that encompasses ecosystem restoration, habitat enhancement, and endangered species protection found in figure S1(b). Rather than valuing these components independently, as is common in existing literature, our proposed approach explicitly recognizes their interdependence and cumulative contribution to overall ecological integrity. Improvements in water quality often simultaneously enhance multiple ecological functions, and treating them separately risks double-counting or, conversely, undervaluing synergistic ecological gains.

Improvements in habitat quality and increases in protected or restored acreage should be valued using appropriate non-market valuation methods. These may include WTP estimates for biodiversity preservation, habitat protection, and endangered species conservation, which reflect not only use values but also existence, bequest, and cultural values held by local and regional communities. Valuation frameworks should also explicitly link ecological improvements to broader social and community benefits.

Enhanced water quality can expand and improve recreational opportunities such as fishing, swimming, boating, and wildlife observation. These benefits can be monetized using WTP estimates for recreation and visitation, with particular attention to seasonal effects during recreation-intensive periods, such as summer months, when marginal improvements in water quality may generate disproportionate benefits. Incorporating seasonality and usage patterns ensures that valuation captures realized benefits rather than hypothetical average conditions.

In addition to direct ecological and recreational benefits, we recommend incorporating secondary economic effects associated with improved water quality. Enhanced environmental amenities are often capitalized into higher residential and commercial property values, particularly in proximity to restored water bodies, wetlands, or improved river corridors. These increases can be quantified through hedonic pricing methods and translated into higher property tax revenues, providing a tangible and recurring fiscal benefit to municipalities. Including these effects helps bridge ecological valuation with local government finance and strengthens the policy relevance of water reuse investments.

Taken together, this integrated valuation approach enables a more comprehensive assessment of water-quality benefits by capturing direct ecological improvements, associated social and cultural values, and downstream economic impacts.

3.3.3. Flood mitigation benefits

To capture the full range of flood mitigation benefits, we recommend adopting a valuation that jointly accounts for impacts at both the municipal and household levels. Flood mitigation measures often generate benefits across multiple scales simultaneously, and evaluating these effects in isolation can substantially understate their total value, as shown in figure S1(c).

At the municipal level, flood mitigation can yield significant cost savings through avoided flood-insurance payouts and reductions in routine and post-event maintenance expenditures. These savings may include lower labor and equipment costs, diminished needs for infrastructure repair and rehabilitation, reduced sediment and debris management, and fewer emergency response interventions. At the household level, valuation should include avoided damage to both structures and contents, encompassing repair and replacement costs for buildings, appliances, vehicles, and personal property. In addition, reductions in flood risk can translate into lower insurance premiums or reduced reliance on flood insurance, particularly in areas where mitigation measures lead to revised risk classifications or lower claims experience [38, 62, 87, 90].

Beyond immediate financial savings, household-level benefits contribute to broader community resilience. By reducing the frequency and severity of flood-related disruptions, mitigation measures can shorten recovery times, maintain property values, and stabilize neighborhoods following extreme events. Integrating these resilience-related benefits into valuation frameworks highlights the role of flood mitigation not only as a damage-avoidance strategy, but also as an investment in long-term social and economic stability.

3.3.4. Overall recommendations and limitations

We emphasize that the proposed framework is intended to prevent the fragmentation of benefits by embedding all parameters such as financial, ecological, social, and hazard-related within a comprehensive cost-benefit analysis. Such integration ensures that direct financial savings, indirect economic effects, ecological improvements, and community-level benefits are all systematically captured.

An illustrative application of the benefit quantification framework is provided in the Supplementary Information using the Wichita Falls, Texas potable reuse case. This example demonstrates how the framework can be applied to a real drought-response setting by identifying relevant water supply, environmental, economic, and social benefit pathways, while excluding pathways such as flood mitigation that are not central to the local reuse objective.

Several limitations should be acknowledged when applying the proposed framework. One key issue relates to transferability, as many valuation parameters, such as WTP estimates and shadow prices, are inherently context-specific. These values often reflect local socio-economic conditions, regulatory settings, and community preferences, so applying them directly across regions without adjustment may yield less reliable estimates. Another concern is the potential for double-counting benefits, particularly given the system's interconnected nature. For instance, improvements in water quality may simultaneously increase recreational opportunities and property values, and without careful separation of these pathways, their effects may be unintentionally counted more than once. The framework should therefore apply a conservative counting rule: each measurable economic indicator leading to a benefit should be counted only once. Monetary estimates that capture overlapping benefit pathways should not be added together unless the overlap has been clearly identified and adjusted for.

For example, WTP estimates for improved water quality may already reflect recreational, aesthetic, ecological, and existence values. Adding separate estimates for recreation benefits or property-value premiums to the same WTP estimate could therefore

double count part of the same benefit. Similarly, hedonic property-value estimates may already capture improvements in water quality, recreation, and flood-risk reduction through changes in housing prices. These estimates should not be added to separate values for the same amenity or risk-reduction benefit unless the benefit pathways are clearly separated. The same principle applies to flood mitigation. Avoided expected flood damages should be treated as the main resource benefit. NFIP claims can be used as a proxy for insured damages or to represent the insurer-program perspective, but they should not be treated as a complete measure of total flood losses. Premium reductions should also be interpreted carefully. If avoided damages have already been counted, premium reductions should be treated as private financial effects rather than additional social benefits.

In addition, the framework may not fully capture dynamic system behavior, as it relies on relatively stable relationships between environmental improvements and economic outcomes, yet these relationships can shift over time due to factors such as climate variability, policy changes, and evolving societal behavior. Similarly, scale differences present a challenge, as some benefits occur at the local level (e.g. ecosystem restoration), while others manifest at broader regional or longer-term scales, making it difficult to integrate them within a single analytical structure. The monetary values, assumptions, and data sources used to estimate benefits are context-specific. Local hydrology, water scarcity, treatment requirements, energy prices, water tariffs, crop types, regulatory standards, property markets, insurance systems, income levels, and public preferences can all affect the magnitude of estimated benefits. Therefore, the framework should be used as a guide for identifying relevant benefit pathways and selecting appropriate valuation methods, rather than as a fixed set of transferable values [64, 123]. When applied outside the original study setting, valuation estimates should be locally calibrated where possible, and any transferred values should be reported with clear assumptions, uncertainty ranges, and sensitivity analysis.

Lastly, the proposed framework provides a structured bridge between benefit identification and monetization as seen in table S1. It organizes water reuse benefits into major pathways and links them to measurable indicators, valuation methods, and aggregation considerations. To apply the framework in a project-level valuation, analysts should define the reuse intervention and the counterfactual condition, identify the relevant benefit pathways, translate each pathway into a physical indicator and unit, select an appropriate valuation method and data source, and determine whether the resulting estimates are additive, overlapping, proxies, or alternative measures of the same benefit. In this sense, the framework supports systematic benefit accounting and valuation-method selection for cost-benefit analysis, while

recognizing that final monetary estimates depend on project-specific hydrologic, engineering, economic, and social data.

4. Conclusion

The systematic review of the literature on water reuse projects reveals several important insights as well as persistent methodological gaps in how the benefits of such projects are assessed. Although many studies emphasize the economic advantages of water reuse most commonly through metrics such as water savings and avoided imported water costs, there remains a limited integration of the broader environmental and social co-benefits that reuse systems generate. In particular, benefits related to water quality improvements, energy savings, reductions in CO₂ emissions, and flood mitigation are often examined in isolation or treated qualitatively, resulting in an incomplete representation of the total value of water reuse. This is especially consequential in regions experiencing water scarcity or hydrologic stress, where the cumulative benefits of reuse may be substantial.

The review also indicates that current investigations disproportionately focus on stormwater as the primary reuse source, while the valuation of other reuse pathways receives comparatively less attention. Furthermore, the enhancement of water quality through reuse—improving ecosystem function, supporting recreational activities, and increasing property values, is rarely assessed within an integrated framework. Similarly, flood mitigation benefits, such as avoided flood damages, reduced insurance costs, and lower municipal maintenance expenditures, are seldom monetized comprehensively, despite their relevance for both household- and community-level resilience.

More broadly, the review highlights that methods for valuing non-market environmental and social benefits remain insufficiently standardized. Additional challenges extend beyond valuation methods alone: there is limited integration of multi-benefit assessment into policy, regulatory, and funding frameworks, and uneven acceptance of existing tools among practitioners and decision-makers. Moreover, the literature is geographically concentrated in Europe and Australia, generating a U.S.-specific knowledge gap due to differences in regulatory environments, climatic conditions, and scarcity challenges. Developing valuation approaches tailored to U.S. contexts while adapting proven models from other regions and sectors would support the creation of more standardized, transparent, and transferable frameworks suitable for California and other states.

To address these gaps, the proposed conceptual framework integrates benefits across three major domains; water supply, water quality, and flood mitigation, within a unified cost-benefit analysis.

This approach expands traditional assessments by incorporating avoided opportunity costs, social costs of carbon reductions, and the broader ecological and economic gains associated with water quality improvements and flood-risk reduction. By synthesizing both direct financial savings and indirect societal and environmental impacts, the framework provides a more holistic and policy-relevant evaluation of water reuse projects. Such integration ensures that both immediate operational benefits and long-term societal outcomes are recognized, supporting more sustainable and equitable water management decisions.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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References

- [1] Joseph N, Ryu D, Malano H M, George B and Sudheer K P 2020 A review of the assessment of sustainable water use at continental-to-global scale *Sustain. Water Resour. Manage.* **6** 18
- [2] Barbier E B 2019 A global crisis in water management (The Water Paradox) (available at: <https://api.semanticscholar.org/CorpusID:204654103>)
- [3] Ghosh S 2022 Global challenge of water crisis *J. Ecol. Nat. Resour.* **6** 1–17
- [4] Diaz M, Arenas A, Schilling K E and Wolter C F 2025 Water balance analysis in Iowa using satellite-based evapotranspiration, precipitation, and streamflow data *Agric. Water Manage.* **322** 109977
- [5] UNESCO World Water Assessment Programme 2024 *Water for Prosperity and Peace* (United Nations Educational, Scientific and Cultural Organization)
- [6] Partlow J 2023 The Colorado River drought crisis: how did this happen? Can it be fixed? Publication by The Washington Post (available at: <https://www.sej.org/headlines/colorado-river-drought-crisis-how-did-happen-can-it-be-fixed>) (Accessed 2024)
- [7] Quench 2023 6 states with water shortages today quench water (available at: <https://quench.culligan.com/blog/states-with-water-shortages/>) (Accessed 2023)
- [8] Crespo D, Nemati M, Dinar A, Frankel Z and Halberg N 2023 Choices magazine 1 economic impacts of climate change on the agricultural sector of the Colorado River basin (available at: www.choicesmagazine.org/choices-magazine/theme-articles/colorado-river-basin/economic-impacts-of-climate-change-on-the-agricultural-sector-of-the-colorado-river-basin#:~:text=This%20means%20that%20as%20water,in%20net%20income%20escalate%20rapidly)
- [9] Booker J F 1995 Hydrologic and economic impacts of drought under alternative policy responses *J. Am. Water Resour. Assoc.* **31** 889–906
- [10] Hamilton P A and Miller T L 2002 Lessons from the National water-quality assessment *J. Soil Water Conserv.* **57** 16A–21A
- [11] Bella J, Sullivan J K, Talley R and Walters R A 2017 Buried no longer: confronting America's Water Infrastructure challenge *Environmental Science Engineering, Political Science* (American Water Works Association [AWWA])
- [12] Shanaghan P E 2012 Assessing drinking water infrastructure need *J. Am. Water Works Assoc.* **104** 14–15
- [13] US EPA 2023 Basic information about water reuse US EPA (available at: <https://www.epa.gov/waterreuse>) (Accessed 2024)
- [14] Tay J-H and Chui P-C 1991 Reclaimed wastewater for industrial application *Waf. Sci. Technol.* **24** 153–60
- [15] Hamdy A 2011 Irrigation with treated municipal wastewater. in
- [16] Jiménez-cisneros B E 2014 3.15 - Water Reuse and Recycling *Water Reuse and Recycling* vol 3 (Comprehensive Water Quality and Purification) pp 296–323
- [17] van Lier J B and Huibers F P 2010 From unplanned to planned agricultural use: making an asset out of wastewater *Irrig. Drain. Syst.* **24** 143–52
- [18] Ghernaout D and Elboughdiri N 2019 Upgrading wastewater treatment plant to obtain drinking water *OAlib* **06** 1–14
- [19] Cotruvo J A and Bell K Y 2014 Need for direct and indirect potable water reuse specifications *J. Am. Water Works Assoc.* **106** 28–30
- [20] Lahnsteiner J, van Rensburg P and Esterhuizen J J B 2018 Direct potable reuse—a feasible water management option *J. Water Reuse Desalin.* **8** 14–28

- [21] Rodríguez C, Van Buynder P, Lugg R, Blair P, Devine B, Cook A and Weinstein P 2009 Indirect potable reuse: a sustainable water supply alternative *Int. J. Environ. Res. Public Health* **6** 1174–209
- [22] Drewes J E, Fox P and Nellor M H 2000 Efficiency and sustainability of soil-aquifer treatment for indirect potable reuse of reclaimed water *Schr.-Reihe Ver. Wasser-, Boden-Lufthyg.* **105** 227–32
- [23] Osipi S R, Secchi A R and Borges C P 2018 Cost assessment and retro-techno-economic analysis of desalination technologies in onshore produced water treatment *Desalination* **430** 107–19
- [24] Boysen B, Cristóbal J, Hilbig J, Güldemund A, Schebek L and Rudolph K-U 2020 Economic and environmental assessment of water reuse in industrial parks: case study based on a model industrial park *J. Water Reuse Desalin.* **10** 475–89
- [25] Anderson J M 2003 The environmental benefits of water recycling and reuse *Water Sci. Technol. Water Supply* **3** 1–10
- [26] Hamilton A J *et al* 2006 Balancing environmental impacts and benefits of wastewater reuse *WSEAS Trans. on Environment and Development* **2** 117–29
- [27] Mitchell V G, Deletic A, Fletcher T D, Hatt B E and McCarthy D T 2007 Achieving multiple benefits from stormwater harvesting *Water Sci. Technol.* **55** 135–44
- [28] Feng W, Liu Y and Gao L 2022 Stormwater treatment for reuse: current practice and future development—a review *J. Environ. Manage.* **301** 113830
- [29] Zhen S, Xun Y, Xu S, Lantian S and Zhou Y 2015 Analysis of plans for rainwater reuse in a residential area in Jiangsu, China *Proc. 2015 Asia-Pacific Energy Equipment Engineering Research Conf. (China) Advances in Engineering Research* (Atlantis Press) pp 367–70
- [30] Martínez-acosta L, López-lambrano A A and López-ramos A 2019 Design criteria for planning the agricultural rainwater harvesting systems: a review *Appl. Sci.* **9** 5298
- [31] Villarreal E L and Dixon A 2005 Analysis of a rainwater collection system for domestic water supply in Ringdansen, Norrköping, Sweden *Build. Environ.* **40** 1174–84
- [32] Ben Brahim-neji H, Ruiz-villaverde A and González-gómez F 2014 Decision aid supports for evaluating agricultural water reuse practices in tunisia: the cebala perimeter *Agric. Water Manage.* **143** 113–21
- [33] Alfarrá A, Kemp-benedict E, Hötzl H, Sader N and Sonneveld B 2011 A framework for wastewater reuse in jordan: utilizing a modified wastewater reuse index *Water Resour. Manage.* **25** 1153–67
- [34] Ait-mouheb N *et al* 2018 The reuse of reclaimed water for irrigation around the Mediterranean Rim: a step towards a more virtuous cycle? *Reg. Environ. Change* **18** 693–705
- [35] US EPA 2023 Water reuse action plan US EPA (available at: <https://www.epa.gov/waterreuse>) (Accessed 2024)
- [36] Water State Revolving Branch C 2021 Integrating water reuse into the clean water state revolving fund (available at: www.epa.gov/waterreuse/water-reuse-action-plan)
- [37] Wang S, Broley W and Mackey E D 2017 Potable REUSE: where we have been and where we are headed *J. Am. Water Works Assoc.* **109** 60–63
- [38] Diringer S E, Shimabuku M, Cooley H and Romano G 2020 Economic evaluation of stormwater capture and its multiple benefits in California *PLoS One* **15** e0230549
- [39] Tortajada C 2021 Water reuse to address water security *Int. J. Water Resour. Dev.* **37** 581–3
- [40] US EPA 2024 *Maps of States with Water Reuse Regulations or Guidelines* US EPA Preprint at (U.S. Environmental Protection Agency)
- [41] Maseeh G P, Russell C G, Villalobos S L, Balliew J E and Trejo G 2015 El Paso's advanced water purification facility: a new direction in potable reuse *J. Am. Water Works Assoc.* **107** 36–45
- [42] Guerrero P B 2016 Direct potable reuse In El Paso, Texas: pioneering water resource sustainability in a large urban utility (available at: https://digitalcommons.utep.edu/open_etd/)
- [43] Hejase C A *et al* 2021 Opportunities for treatment and reuse of agricultural drainage in the United States *ACS ES&T Eng.* **2** 292–305
- [44] Hastie A G, Otrubina V V and Stillwell A S 2023 Identifying opportunities for nonpotable water reuse based on potential supplies and demands in the United States *ACS ES&T Water* **3** 311–21
- [45] Hastie A G, Otrubina V V and Stillwell A S 2022 Lack of clarity around policies, data management, and infrastructure may hinder the efficient use of reclaimed water resources in the United States *ACS ES&T Water* **2** 2289–96
- [46] Harris-lovett S R, Binz C, Sedlak D L, Kiparsky M and Truffer B 2015 Beyond user acceptance: a legitimacy framework for potable water reuse in California *Environ. Sci. Technol.* **49** 7552–61
- [47] Moher D, Liberati A, Tetzlaff J, Altman D G and Group P Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement (available at: www.annals.org)
- [48] McCandless Rob 2024 Advancing water reuse in the US states develop potable reuse regulations (available at: www.wateronline.com/doc/advancing-water-reuse-in-the-u-s-states-develop-potable-reuse-regulations-0001)
- [49] Brown T C 2003 Introduction to Stated Preference Methods *A Primer on Nonmarket Valuation* pp 99–110
- [50] Bateman I J, Keeler B, Olmstead S M and Whitehead J 2023 Perspectives on valuing water quality improvements using stated preference methods *Proc. Natl Acad. Sci. USA* **120** e2217456120
- [51] Scarpa R, Chilton S M, Hutchinson W G and Buongiorno J 2019 Valuing the recreational benefits from the creation of nature reserves in irish forests standard-nutzungsbedingungen (available at: <https://hdl.handle.net/10419/154965>)
- [52] Morley J, Guilfoos T, Mehrabi S, Wang Y and Ikuma K 2025 Willingness to pay for water reuse in small communities in the United States *Water Resour. Res.* **61** e2025WR040116
- [53] Loomis J 2011 What's to know about hypothetical bias in stated preference valuation studies? *J. Econ. Surv.* **25** 363–70
- [54] Loomis J 2014 Strategies for overcoming hypothetical bias in stated preference surveys *J. Agric. Resour. Econ.* **39** 34–46
- [55] Gorno L 2019 Revealed preference and identification *J. Econ. Theory* **183** 698–739
- [56] Michael H J, Boyle K J and Bouchard R 2000 Does the measurement of environmental quality affect implicit prices estimated from hedonic models? *Land Economics* **76** 283–98 (available at: <https://ideas.repec.org/a/uwp/landec/v76y2000i2p283-298.html>)
- [57] Hartman R S 1989 Hedonic methods for evaluating product design and pricing strategies *J. Econ. Bus.* **41** 197–212
- [58] Liu X 2006 The measurement method for the travel time cost and the analysis of its influencing factors. in *Road Traffic Saf.* **6** 19–22
- [59] Herriges J and Kling C 2019 Revealed preference approaches to environmental valuation volumes I and II *Revealed Preference Approaches to Environmental Valuation Volumes I and II* (Routledge) pp 1–595
- [60] Friedrich E, Pillay S and Buckley C A 2007 The use of LCA in the water industry and the case for an environmental performance indicator (available at: www.wrc.org.za)
- [61] Ghimire S R, Johnston J M, Ingwersen W W and Hawkins T R 2014 Life cycle assessment of domestic and agricultural rainwater harvesting systems *Environ. Sci. Technol.* **48** 4069–77
- [62] Molinos-Senante M, Hernández-Sancho F and Sala-Garrido R 2011 Feasibility studies for water reuse projects:

- economic valuation of environmental benefits *NATO Sci. Peace Secur. C* **106** 181–90
- [63] Rolfe J, Johnston R J, Rosenberger R S and Brouwer R 2015 Introduction: Benefit Transfer of Environmental and Resource Values *Benefit Transfer of Environmental and Resource Values: A Guide for Researchers and Practitioners* vol 14 (*The Economics of Non-Market Goods and Resources*) (Springer) pp 3–17
- [64] Richardson L, Loomis J, Kroeger T and Casey F 2015 The role of benefit transfer in ecosystem service valuation *Ecol. Econ.* **115** 51–58
- [65] Wilson M A and Hoehn J P 2006 Valuing environmental goods and services using benefit transfer: the state-of-the art and science *Ecol. Econ.* **60** 335–42
- [66] Vandenberg T, Poe G L and Powell J 1995 Assessing the accuracy of benefits transfers: evidence from a multi-state contingent valuation study of groundwater quality (<https://doi.org/10.22004/ag.econ.127936>)
- [67] Boyle K J, Kuminoff N V, Parmeter C F and Pope J C 2009 Necessary conditions for valid benefit transfers *Am. J. Agric. Econ.* **91** 1328–34
- [68] Barbier E B, Mensah A C E and Wilson M 2023 Valuing the environment as input, ecosystem services and developing countries *Environ. Resour. Econ.* **84** 677–94
- [69] Faber J H, Marshall S, Brown A R, Holt A, van den Brink P J and Maltby L 2021 Identifying ecological production functions for use in ecosystem services-based environmental risk assessment of chemicals *Sci. Total Environ.* **791** 146409
- [70] White A, Faulkner J W, Conner D, Barbieri L, Adair E C, Niles M T, Mendez V E and Twombly C R 2021 Measuring the supply of ecosystem services from alternative soil and nutrient management practices: a transdisciplinary, field-scale approach *Sustainability* **13** 10303
- [71] Barbier E B 2007 Valuing ecosystem services as productive inputs *Econ. Policy* **22** 177–229
- [72] Villa F, Portela R, Onofri L, Nunes P A L D and Lange G M 2015 Assessing biophysical and economic dimensions of societal value: an example for water ecosystem services in Madagascar *Water Ecosyst. Serv.* **A** 110–8
- [73] De Felice F, Campagiorni F and Petrillo A 2013 Economic and environmental evaluation via an integrated method based on LCA and MCDA *Proc. Soc. Behav. Sci.* **99** 1–10
- [74] Adewumi J R, Ilemobade A A and van Zyl J E 2013 Application of a multi-criteria decision support tool in assessing the feasibility of implementing treated wastewater reuse *Int. J. Decis. Support Syst. Technol.* **5** 1–23
- [75] Goyal K and Kumar A 2020 Multi-criteria analysis to identify the factors affecting water reuse scheme in India *Water Environ. J.* **35** 451–62
- [76] McInnes R J, Smith G, Greaves J, Watson D, Wood N and Everard M 2016 Multicriteria decision analysis for the evaluation of water quality improvement and ecosystem service provision *Water Environ. J.* **30** 298–309
- [77] Halaburka B J, Lawrence J E, Bischel H, Hsiao J, Plumlee M H, Resh V H and Luthy R G 2013 Economic and ecological costs and benefits of streamflow augmentation using recycled water in a california coastal stream *Environ. Sci. Technol.* **47** 10735–43
- [78] Dallman S, Chaudhry A M, Muleta M K and Lee J 2016 The value of rain: benefit-cost analysis of rainwater harvesting systems *Water Resour. Manage.* **30** 4415–28
- [79] Domènech L and Sauri D 2011 A comparative appraisal of the use of rainwater harvesting in single and multi-family buildings of the metropolitan area of barcelona (Spain): social experience, drinking water savings and economic costs *J. Clean. Prod.* **19** 598–608
- [80] Folkens L, Wiedemer V and Schneider P 2020 Monetary valuation and internalization of externalities in German agriculture using the example of nitrate pollution: a case-study *Sustainability* **12** 6681
- [81] De Góis E H B, Rios C A S and Costanzi R N 2015 Evaluation of water conservation and reuse: a case study of a shopping mall in southern Brazil *J. Clean. Prod.* **96** 263–71
- [82] Chen R and Wang X C 2009 Cost-benefit evaluation of a decentralized water system for wastewater reuse and environmental protection *Water Sci. Technol.* **59** 1515–22
- [83] de Simone Souza H H, Paulo P L and Boncz M A 2017 A constructed wetland system for residential greywater reuse: economic feasibility of, and willingness to pay for *Desalin. Water Treat.* **91** 336–48
- [84] Declercq R, Loubier S, Condom N and Molle B 2020 Socio-economic interest of treated wastewater reuse in agricultural irrigation and indirect potable water reuse: clermont-ferrand and cannes case studies' cost-benefit analysis *Irrig. Drain.* **69** 194–208
- [85] Alfranca O, García J and Varela H 2011 Economic valuation of a created wetland fed with treated wastewater located in a peri-urban park in Catalonia, Spain *Water Sci. Technol.* **63** 891–8
- [86] Wang R and Zimmerman J B 2015 Economic and environmental assessment of office building rainwater harvesting systems in various U.S. Cities *Environ. Sci. Technol.* **49** 1768–78
- [87] Cooley H, Phurisamban R and Gleick P 2019 The cost of alternative urban water supply and efficiency options in California *Environ. Res. Commun.* **1** 042001
- [88] Wang R, Eckelman M J and Zimmerman J B 2013 Consequential environmental and economic life cycle assessment of green and gray stormwater infrastructures for combined sewer systems *Environ. Sci. Technol.* **47** 11189–98
- [89] Nelson N M, Loomis J B, Jakus P M, Kealy M J, von Stackelburg N and Ostermiller J 2015 Linking ecological data and economics to estimate the total economic value of improving water quality by reducing nutrients *Ecol. Econ.* **118** 1–9
- [90] Paola V, Mustafa A A and Giacomo Z 2018 Willingness to pay for recreational benefit evaluation in a wastewater reuse project. Analysis of a case study *Water* **10** 922
- [91] Hurlimann A and McKay J 2007 Urban Australians using recycled water for domestic non-potable use—an evaluation of the attributes price, saltiness, colour and odour using conjoint analysis *J. Environ. Manage.* **83** 93–104
- [92] Loomis J, Kent P, Strange L, Fausch K and Covich A 2000 Measuring the total economic value of restoring ecosystem services in an impaired river basin: results from a contingent valuation survey *Ecol. Econ.* **33** 103–17
- [93] Alcon F, Pedrero F, Martín-ortega J, Arcas N, Alarcon J J and De Miguel M D 2010 The non-market value of reclaimed wastewater for use in agriculture: a contingent valuation approach *Span. J. Agric. Res.* **8** 187–96
- [94] Kontogianni A, Langford I H, Papandreu A and Skourtos M S 2003 Social preferences for improving water quality: an economic analysis of benefits from wastewater treatment *Water Resour. Manage.* **17** 317–36
- [95] Tziakis I, Pachiadakis I, Moraitakis M, Xideas K, Theologis G and Tsagarakis K P 2009 Valuing benefits from wastewater treatment and reuse using contingent valuation methodology *Desalination* **237** 117–25
- [96] Dupont D P 2013 Water use restrictions or wastewater recycling? A Canadian willingness to pay study for reclaimed wastewater *Water Resour. Econ.* **1** 61–74
- [97] Menegaki A N, Hanley N and Tsagarakis K P 2007 The social acceptability and valuation of recycled water in Crete: a study of consumers' and farmers' attitudes *Ecol. Econ.* **62** 7–18
- [98] Petousi I, Fountoulakis M S, Stentiford E I and Manios T 2015 Farmers' experience, concerns and perspectives in using reclaimed water for irrigation in a semi-arid region of crete, Greece *Irrig. Drain.* **64** 647–54

- [99] Madi M A, Braadbaart O, Al-sa'ed R and Alaerts G 2003 Willingness of farmers to pay for reclaimed wastewater in Jordan and Tunisia *Water Supply* **3** 115–22
- [100] Lazaridou D, Michailidis A and Mattas K 2019 Evaluating the willingness to pay for using recycled water for irrigation *Sustainability* **11** 5220
- [101] Hurlimann A and McKay J 2005 Contingent valuation by the community of indirect benefits of using recycled water: an australian case study *Water Supply* **5** 95–103
- [102] Frew K N, Peterson M N, Sills E, Moorman C E, Bondell H, Fuller J C and Howell D L 2018 Market and nonmarket valuation of North Carolina's tundra swans among hunters, wildlife watchers, and the public *Wildl. Soc. Bull.* **42** 478–87
- [103] van Eeden L M, Bogezi C, Leng D, Marzluff J M, Wirsing A J and Rabotyagov S 2021 Public willingness to pay for gray wolf conservation that could support a rancher-led wolf-livestock coexistence program *Biol. Conserv.* **260** 109226
- [104] Martínez-espiñeira R 2006 'Adopt a hypothetical pup': a count data approach to the valuation of wildlife *Environ. Resour. Econ.* **37** 335–60
- [105] Kazmierczak R F 2001 Economic linkages between coastal wetlands and habitat/species protection: a review of value estimates reported in the published literature prepared by economic linkages between coastal wetlands and habitat/species protection: a review of value estimates reported in the published literature (available at: www.agecon.lsu.edu/faculty_staff/IntroFacPages/kazmierczak.htm)
- [106] Knoche S and Lupi F 2013 Economic benefits of publicly accessible land for ruffed grouse hunters *J. Wildl. Manage.* **77** 1294–300
- [107] Molinos-Senante M, Hernandez-sancho F and Sala-Garrido R 2013 Tariffs and cost recovery in water reuse *Water Resour. Manage.* **27** 1797–808
- [108] Bellver-domingo A, Fuentes R and Hernández-Sancho F 2017 Shadow prices of emerging pollutants in wastewater treatment plants: quantification of environmental externalities *J. Environ. Manage.* **203** 439–47
- [109] Hernández-sancho F, Molinos-Senante M and Sala-Garrido R 2010 Economic valuation of environmental benefits from wastewater treatment processes: an empirical approach for Spain *Sci. Total Environ.* **408** 953–7
- [110] Molinos-senante M, Hanley N and Sala-garrido R 2015 Measuring the CO₂ shadow price for wastewater treatment: a directional distance function approach *Appl. Energy* **144** 241–9
- [111] Das S and Radhakrishnan K 2019 Multicriteria decision making model of wastewater reuse: a stakeholders' perspective in the context of India *Desalin. Water Treat.* **163** 17–25
- [112] Hadipour A, Rajaei T, Hadipour V and Seidirad S 2016 Multi-criteria decision-making model for wastewater reuse application: a case study from Iran *Desalin. Water Treat.* **57** 13857–64
- [113] Paul M, Negahban-azar M, Shirmohammadi A and Montas H 2020 Assessment of agricultural land suitability for irrigation with reclaimed water using geospatial multi-criteria decision analysis *Agric. Water Manage.* **231** 105987
- [114] Drechsel P, Qadir M and Wichelns D 2015 Wastewater: economic asset in an urbanizing world *Wastewater: Economic Asset in an Urbanizing World* (Springer) (<https://doi.org/10.1007/978-94-017-9545-6>)
- [115] National Academies of Sciences, Engineering, and Medicine 2016 *Using Graywater and Stormwater to Enhance Local Water Supplies: An Assessment of Risks, Costs, and Benefits* (The National Academies Press) p 211
- [116] Price J I and Heberling M T 2018 The effects of source water quality on drinking water treatment costs: a review and synthesis of empirical literature *Ecol. Econ.* **151** 195–209
- [117] Woodward R T and Wui Y-S 2001 The economic value of wetland services: a meta-analysis *Ecol. Econ.* **37** 257–70
- [118] Madison M and Emond H 2008 Stormwater capture reuse, and treatment for multipurpose benefits *World Environmental and Water Resources Congress 2008: Ahupua'a—Proc. World Environmental and Water Resources Congress 2008* vol 316 pp 1–10
- [119] Zabala J A, Dolores de Miguel M, Martínez-paz J M and Alcon F 2019 Perception welfare assessment of water reuse in competitive categories *Water Supply* **19** 1525–32
- [120] Del villar A and García-lópez M 2023 The potential of wastewater reuse and the role of economic valuation in the pursuit of sustainability: the case of the Canal de Isabel II *Sustainability* **15** 843
- [121] Karoly L A 2008 Valuing benefits in benefit-cost studies of social programs *Technical Report. The RAND Corporation* (available at: <https://api.semanticscholar.org/CorpusID:167183445>)
- [122] Costanza R 2006 Thinking broadly about costs and benefits in ecological management *Integr. Environ. Assess. Manage.* **2** 166–73
- [123] Johnston R J, Rolfe J and Zawojcka E 2018 Benefit transfer of environmental and resource values: progress, prospects and challenges *Int. Rev. Environ. Resour. Econ.* **12** 177–266